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Everybody Works But The Vacant Lot: A Tax-Based Solution to Canada's Housing Crisis

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Introduction

Cities concentrate economic activity: just six Canadian cities generate 50% of Canada's GDP, but cities also have the highest concentration of inequality (StatCan, "Income of Families")(StatCan "GDP by CMA"). I argue that a Land Value Tax (LVT) makes urban economic activity more efficient while reducing inequality. LVT abolishes taxes for productive assets like buildings, and instead taxes the land underneath them. Land should be taxed because, as an inherently non-productive asset, its appreciation causes significant harm to renters and first-time homebuyers. A LVT universal dividend also improves social welfare: lower-income well-being is improved by far more than higher-income well-being is reduced. I make this argument in four sections: (i) theoretical advantages; (ii) empirical verification; (iii) land and inequality; (iv) political economy.

Theoretical Advantages

Why Urban Areas and Urban Land Matter

Three mechanisms explain productivity gains from spatial concentration: sharing, matching, and learning (Duranton 11). Sharing refers to the cost savings firms enjoy by joining a larger, denser economic network where nearby firms share specialized services rather than sourcing them independently. Matching refers to how denser labour markets reduce search frictions, improving firm-worker matches and efficiently allocating workers into jobs that pay higher wages. Learning refers to the positive externality of knowledge spillovers through informal networks and observation. These benefits are modelled by the productivity curve $A(N)$ in Figure 1.

Concentration also generates productivity costs. The most significant is rising land and housing prices (Duranton 20). Land is a positional good, meaning its supply is fixed, and as cities grow and incomes rise, residents bid more competitively for central locations, inflating land prices without any corresponding increase in land's productive capacity. Expensive land and other costs (e.g., traffic, crime, noise) are modelled by the cost curve $C(N)$ in Figure 1.

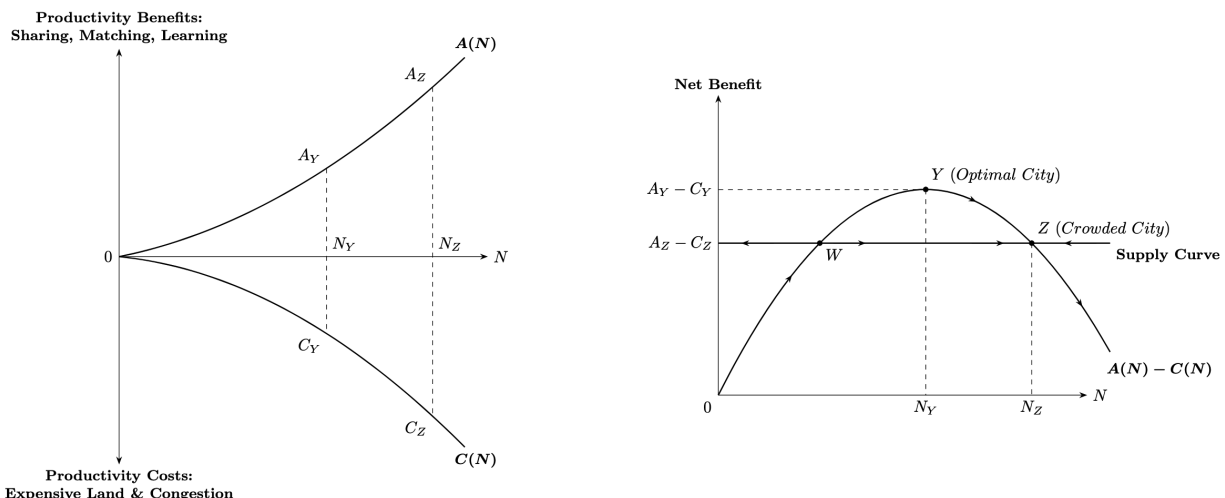


Figure 1

The difference between productivity gains and costs, modelled by the bell-shaped curve $A(N) - C(N)$ in Figure 1, determines a city's net benefit. Net benefit rises initially as productivity gains dominate, peaks at the optimal city size (N_Y), then declines as costs outpace productivity gains. The intersection of the net benefit curve with the supply curve at the crowded city (Z) gives the market equilibrium city size N_Z , which overshoots the social optimum N_Y .

Zoning and privately captured land rents can explain the market's frequent failure to deliver optimal density (Duranton 21). Land rents are returns to landowners above what is required to keep it in current use. Zoning is the body of municipal law dividing areas into districts controlling permissible building types. Properly applied, zoning concerns only compatible land uses (e.g., separating polluting industry from residential zones). However, as Figure 2 illustrates, zoning is frequently exploited by incumbent landowners. They restrict access in order to inflate land rents and maximize their own welfare at the expense of renters. Renters' real wages decline as land rents erode their incomes (Brueckner 42).

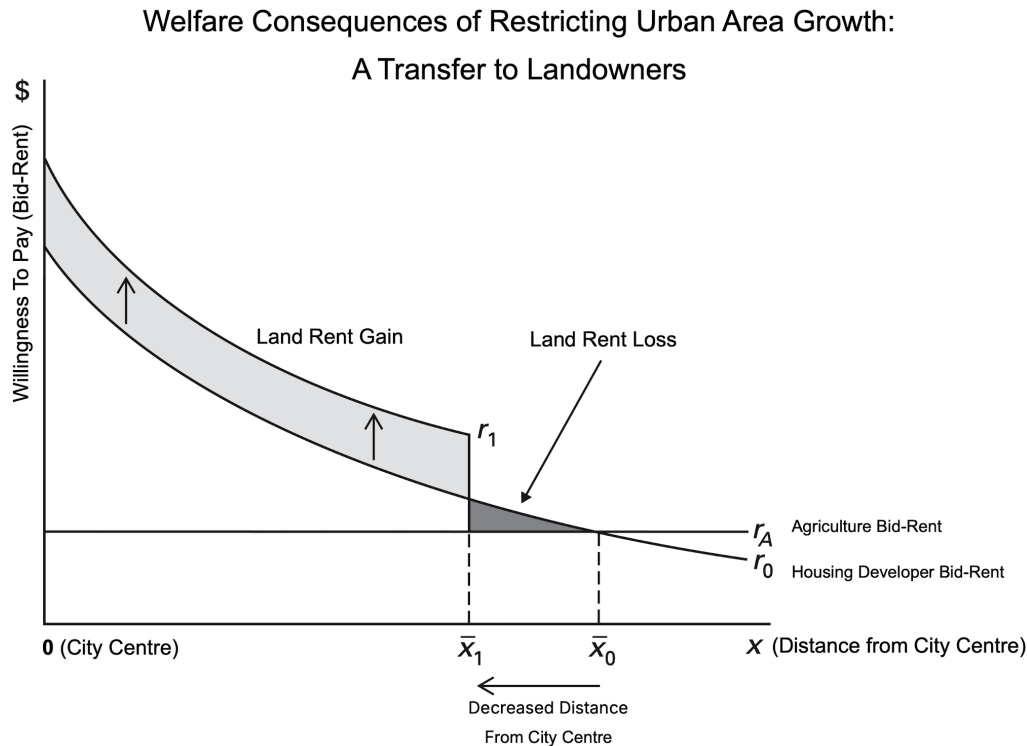


Figure 2

In Figure 2, willingness to pay for housing at any location declines as distance from the city centre increases in proportion to travel costs; this relationship is known as a bid-rent curve. Housing and agriculture compete as alternative land uses, and each location is used by the group with the highest bid-rent curve. The natural city boundary (x_0) forms where the two bid-rent curves intersect. Government-set growth boundaries prevent natural city expansion, thereby moving the city boundary to x_1 . Excess demand then bids rents upwards across the entire urban area. The light-shaded region represents the additional rent incumbent landowners collect due to growth boundaries; the dark-shaded region represents land formerly used by housing and now used by agriculture.

Even without growth boundaries, a growing city continuously transfers a share of its productivity benefits to landowners who did little or nothing to generate them (Duranton 20). Unlike the value added by renovation or new construction, rising land values privately capture the efforts of communal industry: a nearby subway extension, or new businesses moving in raises property

values by virtue of proximity. These privately captured land rents distort incentives toward too little density. Firms and workers bear the full costs of locating in a productive city, but capture only a fraction of the value they generate because expensive housing erodes their real wages (Duranton 20). These problems can be addressed by a LVT paired with zoning reform that promotes intensification (i.e., upzoning).

Land Value Tax (LVT)

A LVT only taxes the unimproved value of land, exempting buildings and structures. This generates three improvements over existing property taxes. First, because land supply is perfectly inelastic (i.e., quantity does not change as price changes), LVT creates no market distortions and cannot be shifted onto tenants. Land's immobility distinguishes it from capital: when a building is included in the tax base, "the owner can save taxes by reducing investments," creating housing scarcity that increases housing costs, effectively shifting the tax onto tenants (ZOE 5). LVT does not trigger this response. Land cannot be withdrawn from the market, and rents (determined by tenants' willingness to pay) are unaffected by taxation.

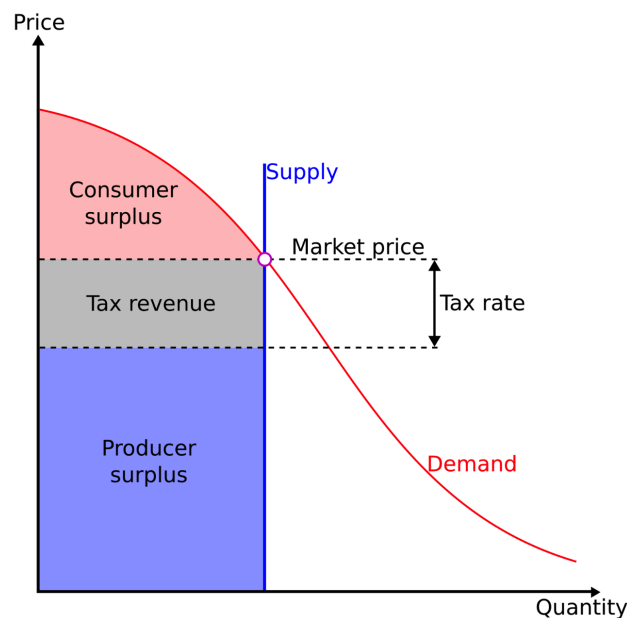


Figure 3: Supply and demand diagram showing that the welfare effects of a tax on a perfectly inelastic supply (land) falls entirely on the producer (landowner) with no welfare loss.

Source: Wikimedia Commons

Second, LVT makes the land component of housing more affordable through capitalization. All assets (including land) can be priced by the present value of future net revenues:

$$V_0 \approx \frac{R}{i}$$

where R is annual rent and i is the discount rate that accounts for the time value of money (e.g., \$100 today is more valuable than \$100 ten years from now)(Williams 77-80). Consider a property whose land rents generate \$15,000/year and whose improvements only generate \$5,000/year. At a 5% discount rate, the total property value is \$400,000: \$300,000 land and \$100,000 improvement.

LVT targets only the property's \$15,000 land rent, leaving its \$5,000 return on improvements untaxed. Because LVT is an annual obligation, it permanently reduces net land rent, and is capitalized directly into land's price:

$$V_1 \approx \frac{R - \tau V_0}{i}$$

where τ is the tax rate and V_0 is the pre-tax land value. When $\tau < i$, land prices fall but remain positive; when $\tau = i$, LVT fully taxes all land rent, and the land's value falls to zero. For example, at a 2% LVT ($\tau < i$), land rent falls to $(\$15,000 - 0.02 \cdot \$300,000) = \$12,000$, while the improvement's rent is unaffected; total property value falls from \$400,000 to \$280,000. How quickly this decapitalization occurs depends on LVT's implementation schedule. But in the long-run, future buyers purchase at the equilibrium reduced price and enjoy that lower cost permanently.

Third, LVT removes the incentives for speculation and disincentives to build inherent in property taxes. Under property taxes, improvements trigger higher assessments, penalizing productive use and encouraging owners to leave land under- or undeveloped. Under LVT, the tax is levied on land value regardless of use, so improvements are never punished, and landowners must generate income sufficient to cover the tax or sell it to someone who will.

Untaxed land rents also make land speculation possible. When credit is available below the expected return on land, borrowers buy land to profit passively from its appreciation and pledge it as collateral for further borrowing without ever increasing the "country's real capital stock"

(Stiglitz, “Inequality” 439). This diverts capital from productive investment, and while the nation’s wealth appears to rise, real wealth falls because the costs for all land-dependent activities increase. The clearest evidence of pure land speculation is the gap between planning approvals and actual construction.¹ Developers buy vacant or underdeveloped land, secure approval to build, but then re-sell, hold, or refinance at the higher post-approval price rather than proceed to construction (StrategyCorp 11). As one planner observed, “the ability to rezone and flip land is built into the provincial planning system” (StrategyCorp 12). LVT would make this cycle prohibitively expensive.

Speculation’s costs fall most heavily on renters. In good times, the value of land as pledged collateral increases, expanding credit availability and encouraging greater land speculation (Stiglitz, “Land and Credit” 16). For renters, rising incomes are largely offset by rising housing costs, leaving them no better off. When overextended land-backed credit inevitably contracts (e.g., run-up to the 2008 financial crisis), incomes fall sharply as the economy enters a recession (Stiglitz, “Land and Credit” 7, 24). Yet rents remain largely fixed and fiscal support flows primarily to homeowners. Following the 2008 financial crisis, for instance, Ontario homeowners received approximately 94% of government housing subsidies, only 6% flowed to renters (Clayton 1). As a result, Figure 4 shows how rent and renters’ income after taxes have diverged since the 1980s declining interest-rate environment (Peak: 19.10% (1981)).

¹For a visual guide to this gap see <https://www.stephenvelasco.com/>

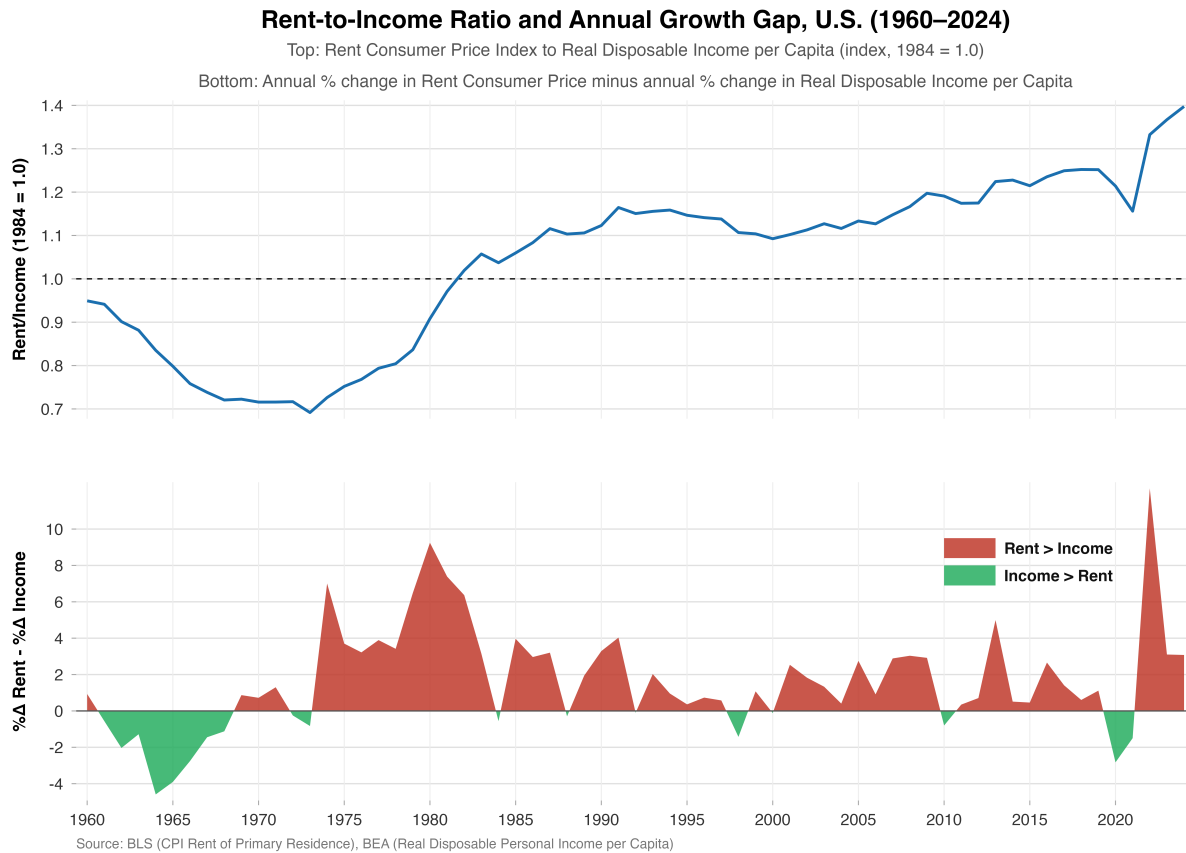
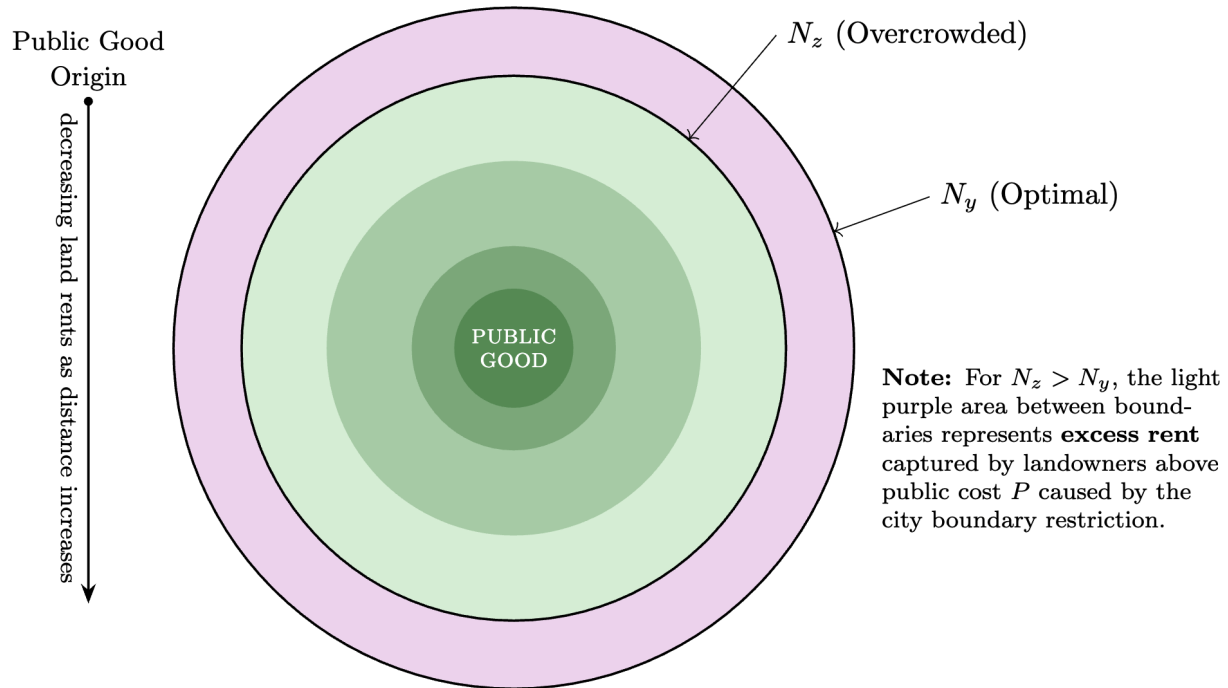


Figure 4

Henry George Theorem of Local Public Finance

Arnott and Stiglitz’s Henry George Theorem (HGT) provides a useful model to determine the optimal rate of land rent recapture. It states that when the government creates a public good, the surrounding land becomes more desirable, and competition for proximity increases aggregate land values by at least as much as the cost of production² (Arnott 472). A well-functioning city (N_y) can thus finance itself by recapturing the land rents its improvements generate. For overcrowded cities (N_z), land rents increase by more than the cost to make the public good (Arnott 478). Figure 5 visualizes this relationship between proximity to public goods and land rents.

² HGT is not universal and only holds under the following conditions: the economy must be large enough; the spatial distribution of economic activity must be Pareto optimal (i.e., no one can be made better off without making someone else worse off, prevented by growth boundaries); and the opportunity cost of land must be uniform along the entire city boundary. In a city like Toronto or Vancouver, it is unlikely that all of these conditions would hold, so HGT may provide “too much or too little tax revenue” (Arnott 477, 485, 498)



$$\int_0^{\text{boundary}} R(t) dt = P$$

land rent (distance from public good) = cost to make public good

Figure 5

At the centre is the public good, and concentric rings form a gradient of decreasing willingness to pay for land as distance from the public good increases in proportion to the travel costs necessary for access. For example, a resident living next to a park enjoys it at no cost, while a resident living farther away must incur a travel cost for the same access. The outer boundary N_y is the optimal unconstrained city, where total land rents exactly equal the public good's production cost. The inner boundary N_z is the crowded city limited by zoning and growth boundaries. The purple ring between the two boundaries represents excess rents that, in an unconstrained city, would spread across a larger land base and aggregate to perfectly finance the public good. Under a growth boundary, however, excess rents capitalize into the constrained land supply, increasing land rents by more than the public good's cost and generating surplus LVT revenue.

This section has shown that LVT establishes efficient incentives to redirect capital away

from speculation and toward productive activities. When paired with upzoning, LVT expands its tax base to accelerate intensification and improve urban amenities. The following section tests these conclusions against Canadian evidence.

Empirical Verification

BCA Methodology

BC Assessments (BCA) values properties by separating them into land and improvements to land (i.e., buildings and structures). Land is valued according to a property's Highest and Best Use (HBU) as if vacant; improvements are then valued as independent contributions to that as-if vacant land. HBU is the “reasonably probable use” of vacant land that must be: (i) legally permitted by zoning or other land use regulations; (ii) physically capable of supporting the use; (iii) financially feasible; and (iv) maximally productive among all feasible uses (HBU Policy 2-3). For example, a parcel in an industrial zone cannot be assessed as a condo site, but a parcel in a residential-inclusive zone may be assessed as a fourplex site if the other criteria are satisfied. Contributory improvement value depends on whether the improvement generates income. If it does (e.g., a rental unit), contributory value is the present value of net operating income over the holding period minus end-of-period costs such as demolition (HBU Policy 6)³. If it does not (e.g., vacant building), contributory value is the market-adjusted reconstruction cost minus depreciation and end-of-period costs (HBU Policy 5)⁴.

Vancouver Case Study

My analysis uses Vancouver property tax reports from 2006 to 2026 to observe macro trends, and BCA parcel-level data for 224,597 properties in 2025 for parcel-level analysis. Two results verify my theoretical argument developed earlier.

³Consider a rented house generating \$20,000 annually (NOI) expected to be redeveloped for apartments in 5 years with demolition that costs \$40,000. Discounting the NPV of NOI and demolition at 6%, the NOI is worth roughly \$84,000 and the future demolition cost is worth roughly \$30,000 today. Therefore, the contributory improvement value is approximately \$54,000 (\$84,000 – \$30,000).

⁴Consider a bungalow on a newly rezoned lot with the same five-year redevelopment plan. The house has over its many years of use accrued age and wear, and has a depreciated replacement value of \$60,000. Demolition costs the same PDV \$30,000, meaning the bungalow's contributory improvement is \$30,000.

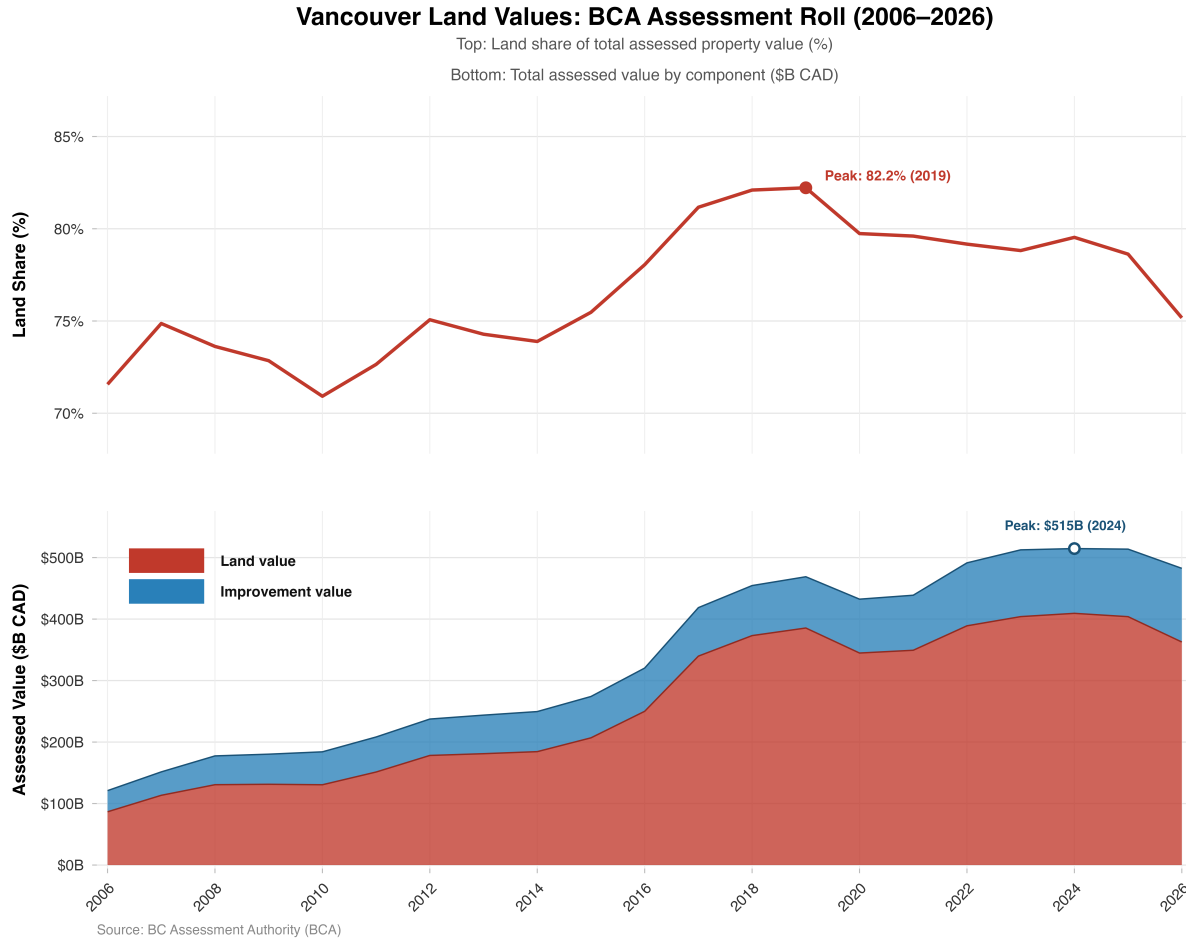


Figure 6

First, as Figure 6 shows, land dominates Vancouver’s total property value and has grown faster than contributory improvements. Aggregate land values rose from approximately \$90 to \$363 Billion (7.42% growth rate), while aggregate improvement values only rose from approximately \$34 to \$120 Billion (6.44% growth rate). Vancouver’s land share of housing (80%) is 30 percentage points greater than the national average (50%)(StatCan “NBSA”). This confirms the welfare transfer predicted by Figure 2: in a highly desirable city with constrained land supply, excess demand raises aggregate land values, and transfers wealth to incumbent landowners.

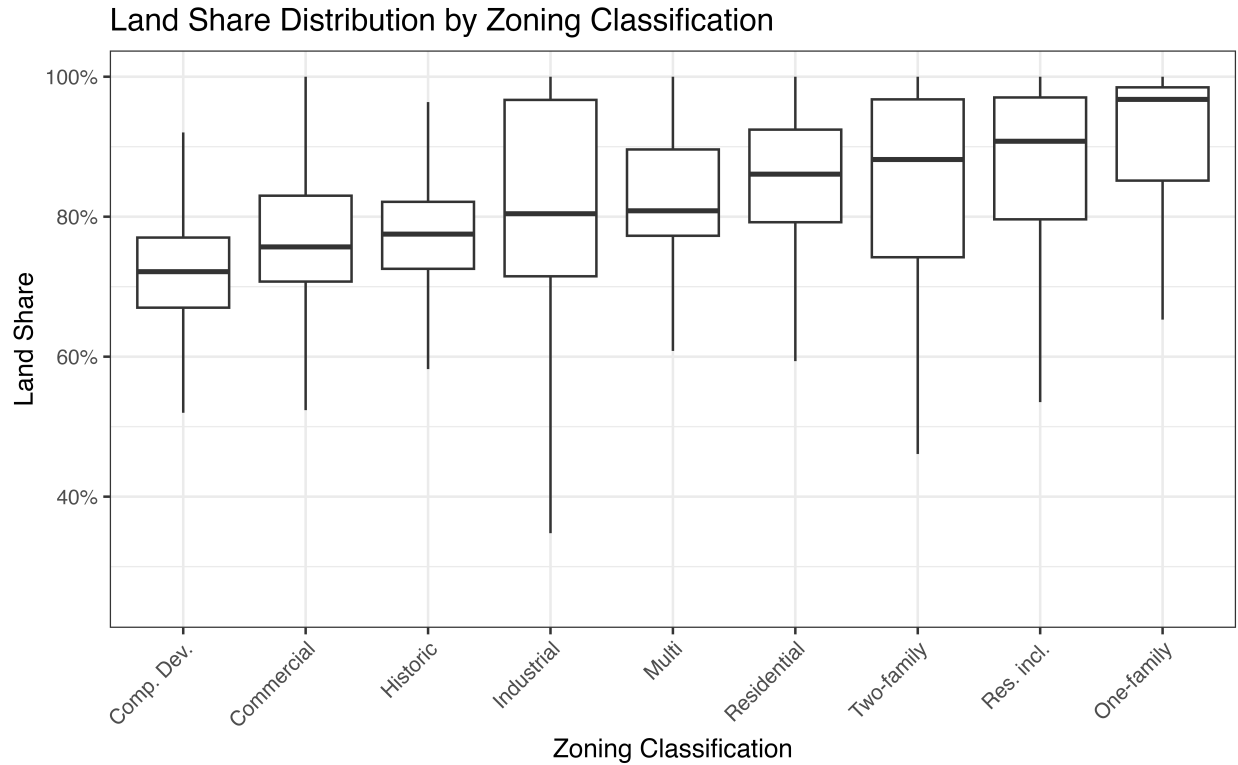


Figure 7

Second, Figures 7 and 8 show a clear inverse relationship between permitted density and land share. Single-family residential (85%) and low-density Residential-Inclusive (87%) zones are the largest by zoned area and most land-concentrated because existing structures contribute minimally relative to the HBU of the underlying land. Commercial (77%) and comprehensive developments like condominiums (72%) occupy far less zoned area, yet are the least land-concentrated because they use the land most efficiently. The zones that use land least efficiently are those where LVT would impose the greatest new obligation and therefore exert the strongest corrective pressure toward more productive use over time.

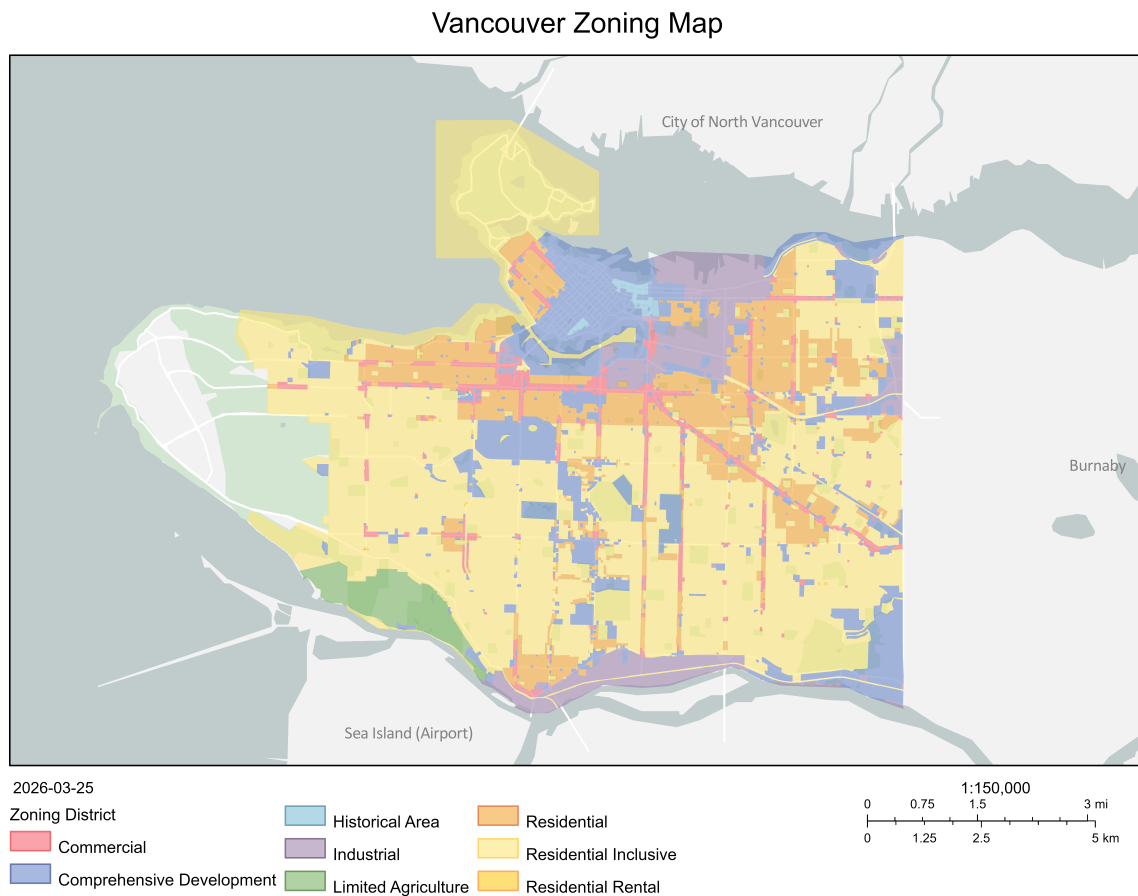


Figure 8

Upzoning

To most effectively correct inefficient land use, LVT must be paired with zoning reform. In 2006, Ontario introduced density and intensification targets ostensibly aimed at reducing suburban sprawl, but with the “underlying policy mechanism to make lower-density forms... more costly, thereby encouraging more compact [development]” (Lorius 97). Under competitive conditions, a home’s market price should approximately equal land plus construction costs. When housing costs persistently exceed reconstruction costs, the difference is an implicit “zoning tax” that has increased over time, as visualized by Figure 9 below.

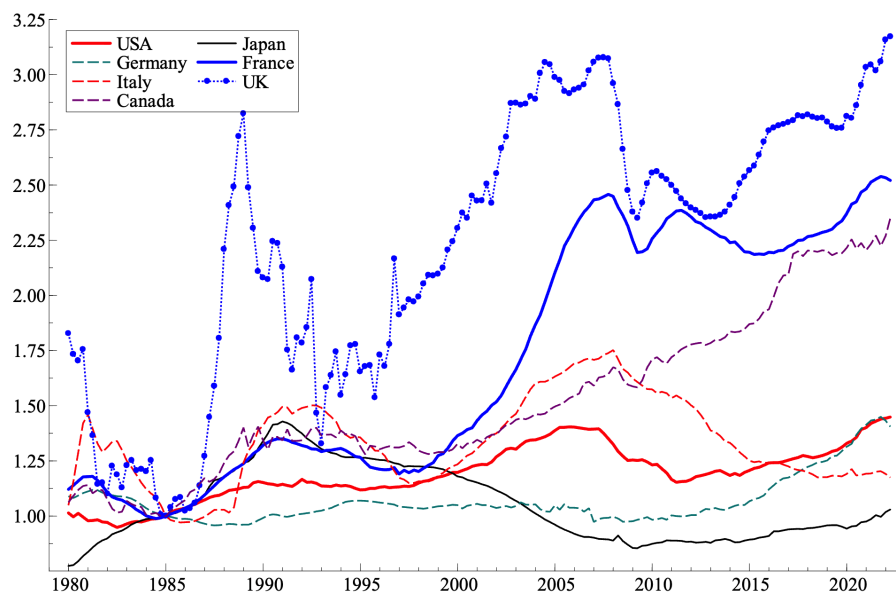


Figure 9: House price-to-construction cost ratio, 1980–2022. Persistent ratios above 1.0 reflect an implicit “zoning tax.” Source: Muellbauer (2023).

Restrictive zoning suppresses HBU and narrows LVT’s tax base. Upzoning raises HBU to accelerate intensification incentives: a landowner taxed on permitted use has far stronger improvement incentives when “permitted use” refers to a twelve-unit apartment building rather than a detached house. Moretti and Duranton estimated upzoning’s efficiency gains and found that “relaxing planning regulations in the three most productive US cities to the median level might generate [GDP] gains of about 8 percent”, primarily through worker reallocation from low- to high-productivity cities currently blocked by unaffordable housing (Duranton 21).

Optimizing Distributional Outcomes

Ownership & LVT Incidence Model

LVT revenue can be spent in three ways: (a) revenue-neutral replacement of existing municipal taxes, (b) expansion of public amenities by HGT, or (c) a household dividend from net revenues. The first two options immediately reduce inequality modestly. Beneficial secondary effects may follow from reduced tax-shifting (revenue neutral) and improved non-car mobility

that disproportionately benefits the poor (public amenities). Only LVT as a universal dividend immediately reduces inequality.

Land ownership is strongly positively correlated with wealth. In Canada, only 24.5% of the poorest 20% were homeowners versus 88.2% of the richest 20%. Figure 10 visualizes ownership inequality and its change since 1996 (StatCan “Homeownership”).

Homeownership Rate by Family Income Quintile, Canada (1999-2016)

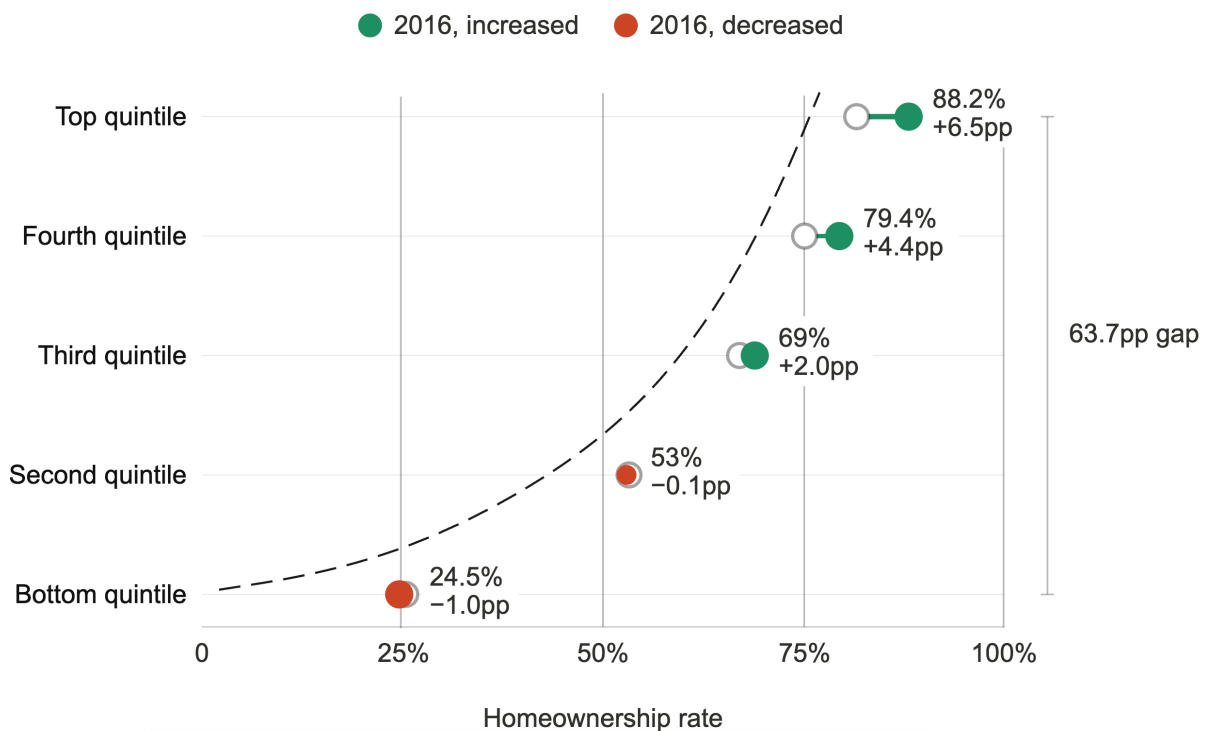


Figure 10

Landowners enjoy significant, tax exempt capital gains on principal residences, while non-owners gain nothing (Eggum and Larsen 25). These rising prices impose higher housing costs on renters, who are disproportionately lower-income and more sensitive to price shocks since housing consumes a larger share of their disposable income. Therefore, to reduce real wage inequality, a LVT dividend, simulated in Figure 11 is necessary:

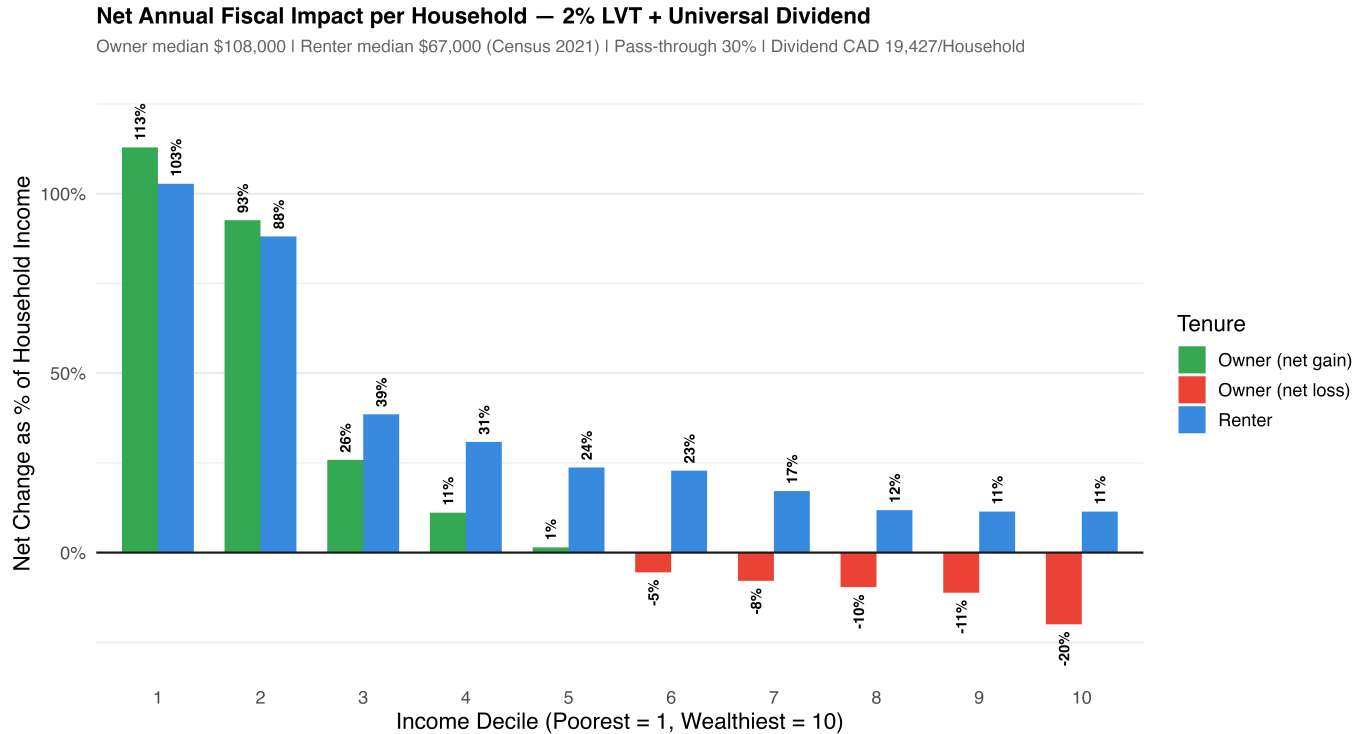


Figure 11

The simulation models the incidence (i.e., who wins or loses from a tax policy) of a 2% LVT on Vancouver households using income deciles from CRA T1 taxfiler microdata and Statistics Canada's 2021 Census owner-renter income distributions, with median incomes of \$108,000 for owners and \$67,000 for renters. LVT revenue is computed from BCA parcel data by applying the city-average land share (79.3%) to total assessed values, and the resulting net revenue over existing levied taxes is redistributed as a uniform per-household dividend.

Ownership rates are modelled as approximately linear by income decile, beginning at 4% in the lowest and rising to 90% in the highest. This assumes that Vancouver's housing market is sufficiently expensive to distinguish itself from national averages and imposes higher entry costs that concentrate ownership among higher-income households. Owner-occupier burden is calculated as each decile's median property value (robust to outliers) multiplied by the city average land share (79.3%) and 2% LVT rate. For renters, a conservative 30% tax shifting incidence is applied to the LVT obligation on a city-median rental property (\$1,170,000), yielding a uniform

annual rent increase of \$5,569. Although LVT incidence is theoretically zero, this is only true in the long-run. Some short-run passthrough is plausible in Vancouver's supply-constrained market, where landlords have limited competition and renters have few close substitutes. Net fiscal impact is then computed for owners as LVT dividend plus tax-shifting rent collection (if landlord) minus LVT obligation; for renters, LVT dividend minus tax-shifting rent increase. All impacts are expressed as a share of disposable income.

The results are strongly progressive: lower-income well-being is improved by far more than higher-income well-being is reduced. The owner's break-even property value is \$1,215,608, meaning that the bottom 60% of property owners would benefit. In real terms, this means that households earning up to \$107,000 would benefit, and the poorest 20% of households would see their incomes double.

Political Economy & Criticisms

The two most significant challenges facing LVT are how it will be implemented and overcoming resistance from wealthy landowners enriched by the status quo.

Implementation

If LVT were suddenly introduced at full rate, the land component of housing would immediately decapitalize. This is a problem for the roughly 66% of Canadians who are homeowners, and especially for the 40% of homeowners who tend to be younger, first-time buyers with a mortgage (FCAC). LVT would make these households owe more on their homes than their homes are worth, as it would immediately devalue the land that serves as collateral on their mortgages, while the debt they carry is unaffected. Sudden implementation is therefore unworkably regressive.

A phase-in approach avoids this. The target rate can be derived from land's local growth rate, introducing LVT such that its annual decapitalization effect exactly offsets land's appreciation. Zero growth would hold land's value constant, giving vulnerable homeowners and the financial sector time to adjust. This transitional period is necessary because approximately 20% of Canadian

bank assets are residential mortgages, and many other assets (e.g., financial derivatives) are indirectly tied to the assumption that land will not decapitalize (Abdelrahman 24). In the long run, however, it is unclear whether banks would be harmed (and may even benefit), as real estate bubbles would become less frequent and less severe.

Reform of existing property taxes is also long overdue. In Ontario, MPAC assessments remain frozen at 2016 values. Older homes are systematically underassessed, while homes that are now a decade old continue to be assessed as if they were brand new. This effectively shifts the property tax burden onto new buyers, who already bear the largest entry costs. LVT, or reform in its direction, would redress this by at least requiring annually updated assessments to undo the freeze.

Political Economy

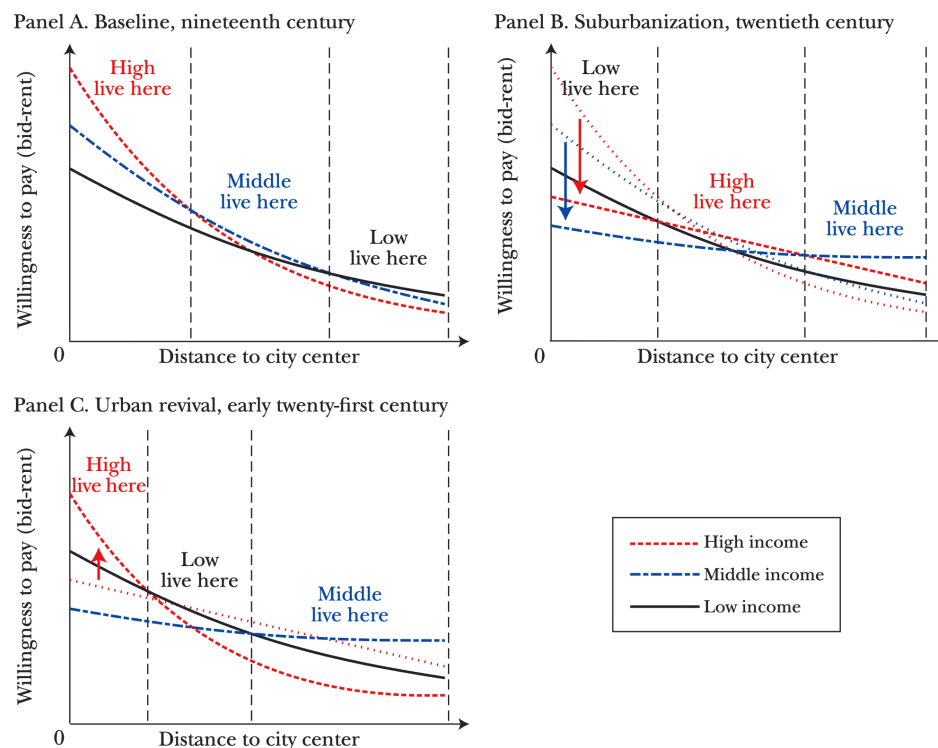


Figure 12: Sorting Patterns by Income in the Monocentric City Model. Source: Couture and Handbury (2023)

Wealthy landowners will inevitably resist LVT if advanced as a purely redistributive policy. LVT becomes politically feasible when framed as the optimal means to provide the amenities that wealthy households have returned to cities to enjoy. As Figure 12 shows, wealthy households have returned to cities to reduce commuting costs and access improved public amenities such as schools and local services (Couture 30, 39). Black (1999) studied 22,679 single-family homes in the Boston area and found that a 5% increase in elementary test scores raised property values on the opposite sides of attendance district boundaries by 2.1% (Black 578). These amenities need not exclusively serve the wealthy. Researchers in Islamabad, Pakistan, found that proximity to slums within 500 meters reduced nearby rents by almost 10% and that resolving those conditions would raise rents within a radius of 1 km by approximately 12.4% (Hussain 1, 10). Universally welfare-improving public investments are currently undersupplied because their gains are privately captured as land rents. LVT corrects this by recapturing those rents to finance the interventions that generated them.

Conclusion

Land Value Tax (LVT) makes urban economic activity more efficient while reducing inequality. LVT abolishes taxes on productive assets like buildings and instead taxes the land underneath them. Land should be taxed, because as an inherently non-productive asset, its appreciation transfers urban productivity gains from workers to landowners, making land speculation possible. If phased in over time, LVT would reduce land prices slowly to allow for banks and first-time buyers to adjust. When LVT revenues are spent as a universal dividend, its outcomes are strongly progressive: owners of property worth less than \$1.2 million benefit, while renters gain uniformly. LVT therefore reduces inequality: lower-income well-being is improved by far more than higher-income well-being is reduced.

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